

Isolectin IB₄ Staining of Whole Mount Adipose Imaging Protocol

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Tested Products:

Isolectin GS-IB₄ Alexa Fluor 488 Conjugate (CAT #I21411, ThermoFischer)
Isolectin GS-IB₄ Alexa Fluor 568 Conjugate (CAT #I21412, ThermoFischer)
Isolectin GS-IB₄ Alexa Fluor 594 Conjugate (CAT #I21413, ThermoFischer)
Isolectin GS-IB₄ Alexa Fluor 647 Conjugate (CAT #I32450, ThermoFischer)

How to Make HEPES Buffer for Dissolving Isolectin:

1. Start 200ml milli-Q water
2. Add 0.074g Calcium Chloride Dihydrate ($\text{CaCl}_2 \cdot 2 \text{H}_2\text{O}$)
Final conc. 1mM
Mol. Wt. 147.0g/mol
3. Add 4.500g Sodium Chloride (NaCl)
154 mOsm/L of Na⁺
Mol. Wt. 58.44g/mol
4. Add 2.979g 1M HEPES (1M = 12.5ml of stock solution)
Final Conc. 25mM
Mol. Wt. 238.3g/mol
5. Fill with milli-Q H₂O to 500ml
6. Check that pH is 7.4, add HCl or NaOH if needed
7. Store at room temp.

Make 100ml Stock of 5ug/mL Isolectin IB₄ Solution (stored in -20°C)

1. Dilute entire 500ug of Isolectin IB₄ in 100mL of HEPES buffer
2. Store at 4°C for short term storage (a couple months) or at -20°C for long term storage (several months to a year).

Whole Mount Adipose Tissue Staining:

To be conducted **after** squishing, blocking, and antibody steps. See protocol for Whole Mount Subcutaneous White Adipose Tissue Immunostaining for details.

1. Dilute desired amount of 1ug/ml Isolectin IB₄ from 5ug/ml stock using HEPES buffer as diluent.
 - a. Note: Stock concentrations have been used for staining as well with equal results. More dilute concentrations have not been thoroughly tested in adipose.
2. Incubate tissues at room temp overnight on shaker.

- a. Note: Incubation at 4°C results in weak/incomplete staining.
3. Next day wash the tissues in 1X PBS for an hour. Repeat once.
4. Check tissues under the microscope for fluorescence.
5. Move onto mounting protocol.

